

Seminar 4

Probability theory - problems for seminar

1. In a group of three students, each solves a problem with probability $1/3$. What is the probability at least of them solve the problem?
2. What is more likely: a) one 6 in 6 rolls or two 6 in 12 rolls?; b) at least one 6 in 6 rolls or at least two 6 in 12 rolls?
3. We pick a number between 0 and 999 and a computer picks one number at random from that range. What is the probability at least two of our digits will match the computer's number? Note: we include any leading zero, so 017 and 057 have two matching digits.
4. You are going to meet a friend at the airport. Your experience tells you that the plane is late 70% of the times it rains, but is late only 20% of the times when it doesn't rain. The weather forecast that morning calls for 40% chance of rain. What is the probability the plane will be late?
5. A student is taking a multiple-choice test in which each question has four possible answers. She knows the answers to 50% of the questions, can narrow the choices down to two 30% of the times and does not know anything about 20% of the questions. What is the probability she will correctly answer a question chosen at random from the test?
6. From a signpost that says MIAMI two letters fall off. A friendly drunk puts the two letters back into the empty slots at random. What is the probability that the sign still says MIAMI?
7. 5% of men and 0.25% of women are colorblind. What is the probability a colorblind person is a man?
8. Which is the probability that in a room with n people someone in the room has the same birthday as you?
9. A family has 3 children, each of them is a boy or a girl with probability $1/2$. Let A and B be the events
 A : "there is at most one girl"
 B : "the family has children of both sexes"
a) Are A and B independent? b) are A and B independent if the family has 4 children?
10. Suppose we pick 4 balls out of a box with 12 red balls and 8 black balls. What is the probability of the event X : "we get two balls of each color"?
11. A box contains m white balls and n black balls. We extract the balls one by one, without reintroducing them back. Which is the probability to obtain the first white ball at the k -th extraction?